

CIRCUMFERENCE AND AREA OF CIRCLES

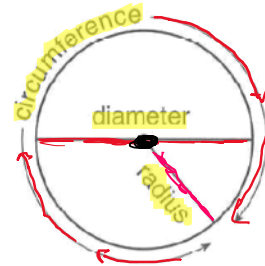
A circle is a set of closed points in a plane. Every point on a circle is the same distance from the center.

2 DIMENSIONS

The distance around a circle is called the circumference. Measuring the circumference of a circle is challenging because you can't use a ruler to measure something round. But you can measure the distance straight across.

The diameter is a line segment that passes through the center of a circle. Drawing a diameter cuts the circle into two equal halves.

Half of the diameter is the radius. The radius is the distance from the center to any point on the circle.



You can approximate circumference using a relationship that exists between the circumference of a circle and its diameter. Whatever the size of the circle, the circumference is always a little more than 3 times the diameter.

$$C \approx 3d$$

$$C/d \approx 3.14$$

$$C/d = \pi$$

a real number, irrational because its decimals NEVER end & NEVER repeat.

In advanced math, we have the formulas for the EXACT value of π even tho we can NEVER find that exact value!!

History and applications

Pi as a limit

The mathematician François Vieta (1540–1603) gave the first theoretically precise expression for π , known as Vieta's formula:

$$\frac{2}{\pi} = \sqrt{\frac{1}{2}} \times \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \times \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}} \times \dots$$

This expresses π as the limit of an infinite product.

John Wallis (1616–1703), who was one of the most influential mathematicians in England in the time just prior to Newton, is also known for the following very beautiful infinite product formula for π :

$$\frac{\pi}{2} = \frac{2 \times 2 \times 4 \times 4 \times 6 \times 6 \times \dots}{1 \times 3 \times 3 \times 5 \times 5 \times 7 \times \dots}$$

Circumference & Area of a Circle

Saturday, August 10, 2024 6:59 PM

The ratio of the circumference to the diameter is called pi, represented by the Greek letter π . The value of π is about 3.14. Pi can be used to write formulas for finding the circumference and area of circles. You should memorize these formulas. They will not be listed on the GED math formula sheet.

Circumference (C) of a Circle

Area (A) of a Circle

$C = \pi d$, where d = diameter

$A = \pi r^2$, where r = radius

$$\pi \approx C/d$$

Note:

① if you are told to use 3.14 in problems, do that.

② if you are told to use the π key on a calculator, then do that.

USING CIRCLE FORMULAS

Example What is the circumference and area of a circle when PQ equals 12 inches?

PQ is the diameter: $d = 12$

The radius is half the diameter: $r = 6$

Use the circumference formula:

$$C = \pi d$$

$$\approx 3.14 \cdot 12$$

ESTIMATE

$$\approx 37.68 \text{ inches}$$

LENGTH

Use the area formula:

$$A = \pi r^2$$

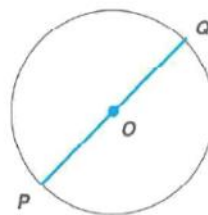
$$\approx 3.14 \cdot 6^2$$

$$\approx 3.14 \cdot 36$$

$$\approx 113.04 \text{ square inches}$$

DITTO

AREA



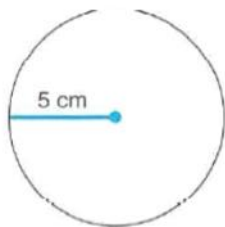
Remember that area is measured in square units. Even though you can't fit squares into a circle, you can estimate how many would fit.

Examples

Saturday, August 10, 2024 7:01 PM

Find the circumference and area of each circle. Round to the nearest tenth.

1.



$$C = d\pi$$

$$d = 2(5) = 10$$

$$\pi \approx 3.14$$

$$C \approx 10(3.14)$$

$$C \approx 31.4 \text{ cm}$$

$$A = \pi r^2$$

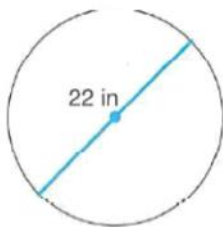
$$r = 5$$

$$\pi \approx 3.14$$

$$A \approx 3.14(25)$$

$$A \approx 78.5 \text{ cm}^2$$

2.



$$C = d\pi$$

$$d = 22$$

$$\pi \approx 3.14$$

$$C \approx 22(3.14)$$

$$\approx 69.08$$

$$C \approx 69.1 \text{ in}$$

$$A = \pi r^2$$

$$r = 11$$

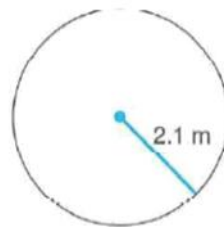
$$\pi \approx 3.14$$

$$A \approx 3.14(121)$$

$$\approx 379.94$$

$$A \approx 379.9 \text{ in}^2$$

3.



$$C = d\pi$$

$$d = 2(2.1) = 4.2$$

$$\pi \approx 3.14$$

$$C \approx 4.2(3.14)$$

$$\approx 13.188$$

$$C \approx 13.2 \text{ m}$$

$$A = \pi r^2$$

$$r = 2.1$$

$$\pi \approx 3.14$$

$$A \approx 3.14(4.41)$$

$$\approx 13.8474$$

$$A \approx 13.8 \text{ m}^2$$

notice: same thing !!

Study Tip

Write the formulas every time you use them.

You will get tired of doing this!

However, it is a great habit because, especially with difficult Math, two things happen:

1. You memorize the formulas, and,
2. It will help you understand the formulas such that you won't have to study for Math tests.

About #2, your instructor does not know what happens in the brain, but something happens to bring understanding!!

Examples, cont'd...

Monday, September 2, 2024 12:11 PM

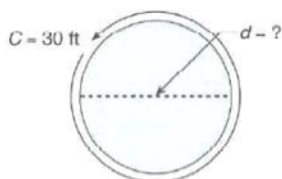
4. Sasha needs to buy one packet of seeds for every 12 square feet in her garden. If the circular garden has a diameter of 18 feet, how many packets of seeds will Sasha need?

$$\begin{aligned} A &= \pi r^2 \\ r &= \frac{1}{2}d \\ &= \frac{1}{2}(18) \\ &= 9 \\ \pi &\approx 3.14 \end{aligned}$$

$$\begin{aligned} A &\approx 3.14(9^2) \\ &\approx 3.14(81) \\ A &\approx 254.34 \text{ sq ft} \end{aligned}$$

$$\begin{aligned} \rightarrow \text{Size 1 Seed Packet} &= 12 \text{ ft}^2 \\ \# \text{ Seed Packets} &= A / 12 \text{ ft}^2 \\ &= 254.34 \text{ ft}^2 / 12 \text{ ft}^2 \\ &= 21.195 \\ \# \text{ Seed Packets} &= 22 \end{aligned}$$

5. Max is planting a circular flower garden in the center of a lawn. He has 30 feet of fencing to go around the garden. To the nearest tenth of a foot, what is the greatest diameter the garden can have without Max buying more fencing?



$$\begin{aligned} C &= 30 \text{ ft} \\ C &= \pi d \\ d &= C / \pi \\ d &\approx 30 / 3.14 \\ &\approx 9.55 \\ d &= 9.6 \text{ ft} \end{aligned}$$

I want to Round down to 9.5 so as to NOT buy more Fencing, but problem said Round to the nearest foot.

6. Cans of frozen juice concentrate often have plastic strips that you pull to remove the lid from the can. The length of the strip is the circumference of the lid plus 0.25 inch. If the radius of the lid is 1.25 inches, what is the length of the plastic strip to the nearest tenth of an inch?



$$C = \text{length} + 0.25$$

$$\begin{aligned} r &= 1.25 \\ d &= 2r = 2(1.25) \\ &= 2.5 \end{aligned}$$

$$\begin{aligned} L &\approx C + 0.25 \approx d\pi + 0.25 \\ L &\approx d\pi + 0.25 \\ &\approx 2.5(3.14) + 0.25 \\ &\approx 7.85 + 0.25 \\ L &\approx 8.1 \text{ in} \end{aligned}$$

AREA OF COMPLEX FIGURES

A complex figure is made from more than one shape. The area of a complex figure can't be found using a single formula. Instead, you have to develop a plan of your own for finding the area.

Some figures can be cut up into smaller pieces. The area is the sum of the areas of these smaller shapes.

ADDING TO FIND AREA

Example What is the area of the figure in square inches?

Step 1

Divide the figure into smaller parts.

Here, the figure is divided into a rectangle and a square.

Step 2

Find the areas of the parts.

Rectangle: $A = 10 \cdot 9$
 $= 90 \text{ sq in}$

Square: $A = 5 \cdot 5$
 $= 25 \text{ sq in}$

Step 3

Find the total area.

$90 + 25 = 115 \text{ sq in}$

They never told us inches !!

$A = L \cdot W$
 $A = (10)(9) = 90 \text{ sq units}$
 $A = (5)(5) = 25 \text{ sq units}$
Total Area
 $= 115 \text{ sq units}$

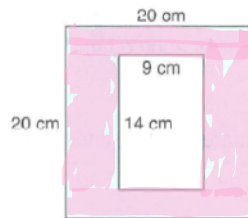
SUBTRACTING TO FIND AREA

Example What is the area of the shaded part of the figure?

Step 1

Divide the figure into smaller parts.

A rectangle is cut out of a square.



Step 2

Find the areas of the parts.

Square: $A = 20 \cdot 20$
 $= 400 \text{ cm}^2$

Rectangle: $A = 14 \cdot 9$
 $= 126 \text{ cm}^2$

Step 3

Find the area of the shaded portion.

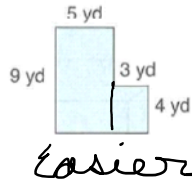
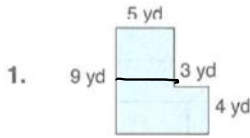
$400 - 126 = 274 \text{ cm}^2$

area of shaded Area
 $=$
 lg area
 minus
 sm area
 $=$
 $(20)(20) - (14)(9)$
 $=$
 $400 - 126$
 $A = 274 \text{ cm}^2$

More Examples

Saturday, August 10, 2024 7:04 PM

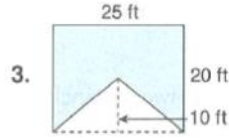
Find the area of each figure.



$$A = (9)(5) + (3)(4)$$

$$= 45 + 12$$

$$A = 57 \text{ yds}^2$$



$$\text{Area}_{\Delta} = \frac{1}{2}bh$$

$$= \frac{1}{2}(25)(10)$$

$$= 125$$

$$\text{Area}_{\square} = bh$$

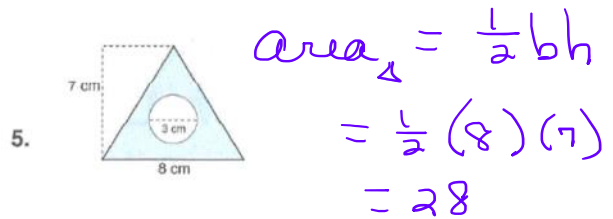
$$= 25(20)$$

$$= 500$$

$$\text{shaded Area}$$

$$= 500 - 125$$

$$A = 375 \text{ ft}^2$$



$$\text{area}_{\Delta} = \frac{1}{2}bh$$

$$= \frac{1}{2}(8)(7)$$

$$= 28$$

$$\text{area}_{\circ} = \pi r^2, r = 1.5$$

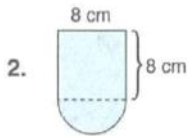
$$\approx 3.14(1.5)^2$$

$$\approx 7.065$$

$$\text{shaded Area}$$

$$\approx 28 - 7.065$$

$$A \approx 20.935 \text{ cm}^2$$



(Hint: This figure contains a half-circle.)

$$\text{Area}_{\square} = s^2$$

$$= (8)^2$$

$$= 64$$

$$\text{area}_{\circ} = \frac{1}{2}\pi r^2$$

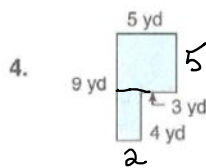
$$\approx \frac{1}{2}(3.14)(4^2)$$

$$\approx 25.12$$

TTL Area

$$A \approx 64 + 25.12$$

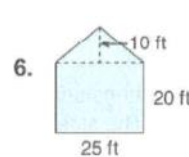
$$A \approx 89.12 \text{ cm}^2$$



$$A = (5)(5)$$

$$+ (2)(4)$$

$$A = 33 \text{ yds}^2$$



$$A = (25)(20)$$

$$+ \frac{1}{2}(25)(10)$$

$$= 500 + 125$$

$$A = 625 \text{ ft}^2$$

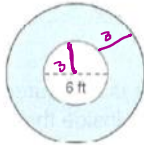
More Examples, cont'd...

Monday, September 2, 2024 12:12 PM

Solve.

7. A circular fountain has a 3-foot walkway surrounding it. What is the area of the walkway?

Round to the nearest tenth.



Large circle

$$\begin{aligned} r &= 6 \\ A &= \pi r^2 \\ &= 3.14 (36) \\ &= 113.04 \end{aligned}$$

Small circle

$$\begin{aligned} r &= 3 \\ A &= \pi r^2 \\ &= 3.14 (9) \\ &= 28.26 \end{aligned}$$

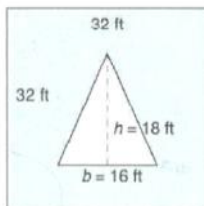
Area of the walkway

$$\begin{aligned} A &= 113.04 - 28.26 \\ &= 84.78 \end{aligned}$$

$$A = 84.8 \text{ ft}^2$$

Do NOT ROUND UNTIL the LAST STEP!!!

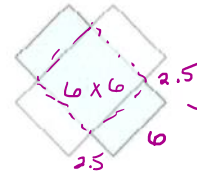
9. The shaded area represents a walkway around a triangular garden. What is the area of the walkway?



$$\begin{aligned} A &= (32^2) - \left[\frac{1}{2} (16) (18) \right] \\ &= 1024 - 144 \end{aligned}$$

$$A = 880 \text{ ft}^2$$

8. An X is formed by drawing one 11-by-6-inch rectangle on top of another. What is the area in square inches of the X?



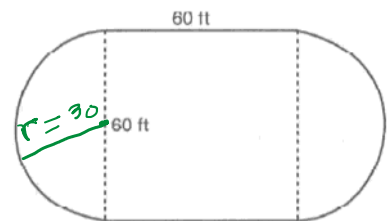
there are 4 of these.

$$A = 4 (2.5)(6) + 6^2$$

$$A = 96 \text{ in}^2$$

nice problem!!

10. An event planner has the following space to create a dirt bike course. To the nearest square foot, what is the area of the space?



$$\begin{aligned} A &= \text{circle} + \text{square} \\ &= 3.14 (30^2) + 60^2 \\ &= 3.14 (900) + 3600 \\ &= 2826 + 3600 \end{aligned}$$

$$A = 6426 \text{ ft}^2$$

11. **Investigate** The figure below is $\frac{3}{4}$ of a circle.



- a. Find the area of the figure.
- b. If a square is drawn with the same area, what would a side of the square measure to the nearest inch?

estimate : 14 in
because $14^2 = 196$

$$\begin{aligned} a) A &= \frac{3}{4} \pi r^2 \\ &= \frac{3}{4} (3.14) (9^2) \end{aligned}$$

$$A = 190.755 \text{ in}^2$$

b) area of square

$$A = 190.755 = s^2$$

$$s = \sqrt{190.755}$$

$$s \approx 13.8 \text{ in}$$